

OpenDaylight server, and OpenDaylight will automatically create a *br-int* bridge.

```
[root@vinoth ~]# ovs-vsctl show
9e3b34cb-fefc-4br4-828s-084b3e55rtfd
Manager "tcp:192.XXX.X.XXX:XXXX"
Is_connected: true
Bridge br-int
Controller "tcp:192.XXX.X.XXX:XXXX"
fail_mode: secure
Port br-int
Interface br-int
ovs_version: "2.1.3"
```

6. Configuring *ml2_conf.ini* for the OpenDaylight driver

Edit *vi /etc/neutron/plugins/ml2/ml2_conf.ini* in all the required nodes and modify the following configuration. Leave the other configurations as they are.

```
[ml2]
type_drivers = flat,vxlan
tenant_network_types = vxlan
mechanism_drivers = opendaylight

[ml2_odl]
password = admin
username = admin
url = http://<OPENDAYLIGHT SERVER's IP>:8080/controller/nb/v2/neutron
```

7. Configuring the Neutron database

Reset the Neutron database, as follows:

```
$ mysql -uroot -p
$ drop database neutron;
$ create database neutron;
$ grant all privileges on neutron.* to 'neutron'@'localhost'
identified by '<YOUR NEUTRON
PASSWORD>';
$ grant all privileges on neutron.* to 'neutron'@ '%'
identified by '<YOUR NEUTRON
PASSWORD>';
$ exit
$ su -s /bin/sh -c "neutron-db-manage --config-file /etc/
neutron/neutron.conf --config-file
etc/neutron/plugins/ml2/ml2_conf.ini upgrade head" neutron
```

Restart the Neutron server, using the following command:

```
$ service neutron-server start
```

8. Installing the *networking_odl* Python module

Important: At this stage you will get a message stating

“Neutron service failed to start.” Don’t worry; it’s a temporary issue. Since you have enabled OpenDaylight as a *mechanism_driver* but not yet installed the Python module for it, you need to install the *networking_odl* Python module, as follows:

```
$ apt-get install python-pip
$ pip install networking_odl
```

Now, restart the Neutron server and check its status. It should be running without errors.

9. Verifying the integration

We are almost done with integrating OpenStack with OpenDaylight. Now, it’s time to verify it.

Create initial networks in OpenStack and check whether they are reflected in OpenDaylight.

You could create the network using the OpenStack horizon dashboard also. Shown below are the commands to create the same in CLI:

```
$ neutron router-create router1
$ neutron net-create private
$ neutron subnet-create private --name=private_subnet
172.0.0.0/24
$ neutron router-interface-add router1 private_subnet
$ nova boot --flavor <flavor> --image <image id> --nic net-
id=<network id> vinoth-vm1
```

Once the network creation is done, you will be able to see that the same network has been created in OpenDaylight by using the following command:

```
$ curl -u admin:admin http://<OPENDAYLIGHT's SERVER>:8080/
controller/nb/v2/neutron/networks
```

The above command will return the network information which is created in OpenDaylight through OpenStack. If the integration has been done correctly, you should now be able to ping VM. 

References

- [1] <http://www.hellovinoth.com/>
- [2] <http://www.cloudenablers.com/blog/>
- [3] <https://www.opendaylight.org/>
- [4] <http://docs.openstack.org/>

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